AP PRECALCULUS

Unit 1 • Lesson 1.4 • Notes (filled in)

Topic 1.4 • Polynomial Functions and Rates of Change

Notes (filled in)

Learning goals

- Identify the degree, leading term and leading coefficient of a polynomial in any form.
- Locate local and global extrema from a graph, including on closed intervals.
- Identify points of inflection and describe the rate of change between turning points.

Vocabulary. polynomial, degree, leading term, leading coefficient, local (relative) extremum, global (absolute) extremum, point of inflection.

1. Polynomial functions

A polynomial of degree n is $p(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$ with a_n $\neq 0$. $a_n x^n$ is the leading term and a_n the leading coefficient. The degree of a product is the sum of the degrees of the factors.

Example 1

Give the degree and leading coefficient of $p(x) = -2x(x-1)^2(x+3)^3$.

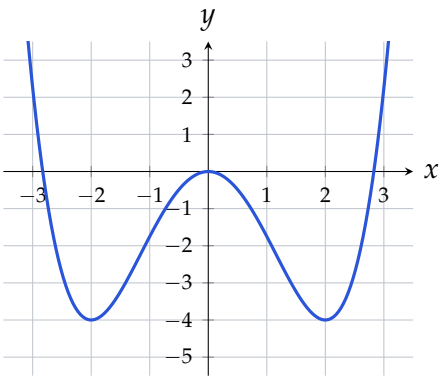
Degree $1+2+3=6$. Leading term $-2x \cdot x^2 \cdot x^3 = -2x^6$: leading coefficient -2.

2. Extrema

Key idea

- A **local maximum** occurs where p changes from increasing to decreasing.
- A **local minimum** occurs where p changes from decreasing to increasing.
- The **global maximum** on an interval is the greatest output value on that interval; on a closed interval it can occur at an endpoint.
- A polynomial of degree n has at most $n-1$ local extrema.

Example 2 • Reading extrema



The graph of $p(x) = \frac{1}{4}x^4 - 2x^2$ is shown. Find all local extrema, the global extrema on $(-\infty, \infty)$, and the global maximum on $[0, 3]$.

Local minima $(-2,-4)$ and $(2,-4)$; local maximum $(0,0)$.
Global minimum value -4; no global maximum (the ends go up).
On $[0,3]$: compare $p(0)=0$, $p(2)=-4$, $p(3)=\frac{81}{4}-18=2.25$. Global maximum 2.25 at $x=3$ (an endpoint).

3. Points of inflection

Key idea
A point of inflection is where the graph changes concavity : the rate of change switches from increasing to decreasing, or the reverse. Between a local maximum and the next local minimum there is at least one point of inflection.

Example 3

$p(x)=x^3-6x^2+9x$ has a local maximum at $(1,4)$, a local minimum at $(3,0)$ and a point of inflection at $(2,2)$. Describe p and its rate of change.

Increasing on $(-\infty,1)$, decreasing on $(1,3)$, increasing on $(3,\infty)$.
The rate of change decreases until $x=2$ (concave down on $(-\infty,2)$) and increases after (concave up on $(2,\infty)$). On $(1,2)$ the function decreases faster and faster; on $(2,3)$ it decreases more and more slowly.

Common mistakes
- Giving the x-value when asked for the extreme *value*, or the reverse.
- Forgetting the endpoints when looking for global extrema on a closed interval.

Check your understanding

1. A degree-6 polynomial has at most 5 turning points.
2. A cubic with no local extrema: $p(x)=x^3$.